Mixed Methods for Two-Phase Darcy-Stokes Mixtures of Partially Melted Materials with Regions of Zero Porosity

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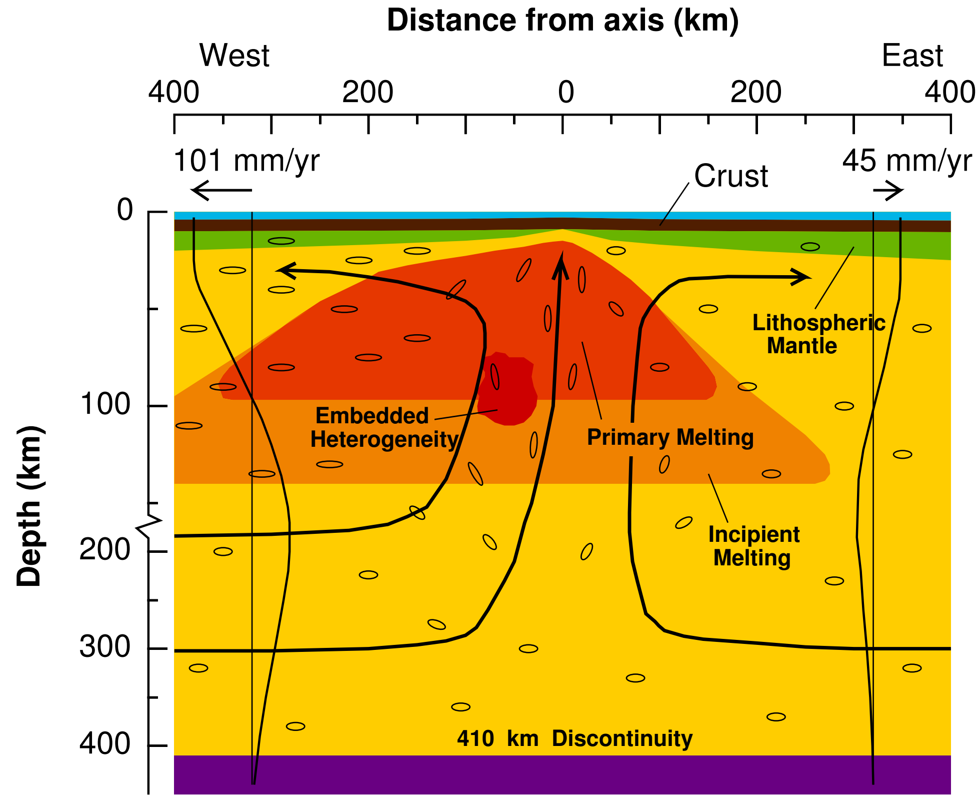


(MELT seismic team, Science, 1998)

Modeling a Mid-Ocean Ridge.

Porosity, $0 \leq \phi(x) \ll 1$ is the volume fraction of fluid melt.

- $\phi = 0 \implies$ solid phase
- $\phi > 0 \implies$ two-phases

Objective. Accurate numerical approximation of the mechanics of a deformable solid matrix that is *possibly* partially melted, as applied to

- the Earth's mantle (rocks);
- glaciers and polar ice sheets (ice).

Goals. Provide a mathematically sound framework for both

1. the model governing the system, and
2. the numerical approximation of the model.

1. The Equations Governing Mantle Dynamics

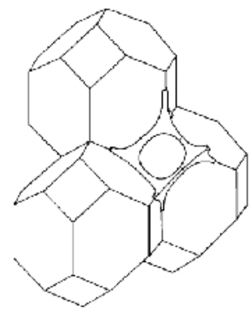
Mixture Theory for Mantle Dynamics
(McKenzie 1984)

Based on mixture theory: Both fluid (f) and solid matrix (s) phases exist at each point of the domain Ω.

For a quantity Ψ, define the mixture variable

$$\overline{\Psi} = \phi\Psi_f + (1-\phi)\Psi_s = \Psi_s + \phi\Psi_r, \quad \Psi_r = \Psi_f - \Psi_s.$$

1. Darcy for fluid melt. Fluid melts between rock crystals, forming a porous medium.
2. Stokes for solid rock. Highly viscous creep of the solid mantle.



3. Thermodynamics and phase behavior. Governs the transient behavior of ϕ. We assume that ϕ is given and consider the mechanics only.

Notation.

$\mathbf{v}_f, \mathbf{v}_s$	Phase velocity	$\mathbf{u}$	Darcy velocity
p_f, p_s	Phase pressure	$\bar{p}$	Mixture pressure
ρ_f, ρ_s	Phase density	$\bar{\rho}$	Mixture density
μ_f, μ_s	Phase viscosity	$\mathbf{g}$	Gravity vector

$\rho_r = \rho_f - \rho_s$ Relative density $\mathbf{v}_r = \mathbf{v}_f - \mathbf{v}_s$ Relative velocity

1. Darcy's Law.

$$\mathbf{u} = \phi \mathbf{v}_r = \phi(\mathbf{v}_f - \mathbf{v}_s) = -\frac{k_0 \phi^{2+2\Theta}}{\mu_f}(\nabla p_f - \rho_f \mathbf{g})$$

The relative permeability $K(\phi) = \phi^{2+2\Theta} K_0$ for simplicity

2. Conservation of mass

$$\frac{\partial}{\partial t}\Big(\phi\rho_f + (1-\phi)\rho_s\Big) + \nabla \cdot \Big(\phi\rho_f \mathbf{v}_f + (1-\phi)\rho_s \mathbf{v}_s\Big) = 0$$

or

$$\frac{\partial}{\partial t}\bar{\rho} + \nabla \cdot \overline{\rho \mathbf{v}} = 0$$

For constant and equal densities for non-buoyancy terms (or a Boussinesq approximation)

$$\nabla \cdot \bar{\mathbf{v}} = \nabla \cdot (\phi \mathbf{v}_r + \mathbf{v}_s) = 0$$

3. Viscous Stokes equation. Conservation of momentum for

- a highly viscous deformable matrix

$$\boldsymbol{\sigma}_s = -p_s\,\mathbf{I} + \mu_s\Big(2\mathcal{D}\mathbf{v}_s - \tfrac{2}{3}\nabla\cdot\mathbf{v}_s\,\mathbf{I}\Big)$$

  where $\mathcal{D}\mathbf{v}_s = \frac{1}{2}(\nabla\mathbf{v}_s + \nabla\mathbf{v}_s^T)$
- and fluid $\boldsymbol{\sigma}_f = -p_f\,\mathbf{I}$
- as a mixture

$$\nabla\cdot\bar{\boldsymbol{\sigma}} = -\bar{\rho}\mathbf{g}$$

or

$$-\nabla\bar{p} + \nabla\cdot\Big((1-\phi)\mu_s\Big(2\mathcal{D}\mathbf{v}_s - \tfrac{2}{3}\nabla\cdot\mathbf{v}_s\,\mathbf{I}\Big)\Big) = -\bar{\rho}\mathbf{g}$$

4. Compaction

$$p_s - p_f = -\frac{\mu_s}{\phi}\nabla\cdot\mathbf{v}_s,$$

5. Thermodynamics and phase behavior. Governs the transient behavior of ϕ. We assume that ϕ is given and consider the mechanics only.

We have four equations and five primary unknowns:

- $\mathbf{v}_r$ and $\mathbf{v}_s$
- p_f, p_s, and $\bar{p}$

We choose to eliminate p_s from the system (simplifies Stokes).

Pressure potentials. Let

- $q_f = p_f - \rho_f g z$
- $q_s = p_s - \rho_f g z \implies p_f - p_s = q_f - q_s$
- $q = \bar{q} = \phi q_f - (1 - \phi) q_s$

Note that

$$p_f - p_s = q_f - q_s = \frac{1}{1 - \phi}(q_f - q)$$

Darcy's law for the fluid

$$\mathbf{u} + \frac{k_0 \phi^{2+2\Theta}}{\mu_f} \nabla q_f = 0 \qquad \text{(Darcy law)}$$

$$\mu_s \nabla \cdot \mathbf{u} + \frac{\phi}{1-\phi}(q_f - q) = 0 \qquad \text{(conservation/ compaction)}$$

Stokes system for a highly viscous, compressible mixture (matrix & fluid)

$$\nabla q - \nabla \cdot \hat{\boldsymbol{\sigma}}(\mathbf{v}_s) = -(1-\phi)\,\rho_r\,\mathbf{g} \equiv \mathbf{G} \qquad \text{(Stokes)}$$

$$\mu_s \nabla \cdot \mathbf{v}_s - \frac{\phi}{1-\phi}(q_f - q) = 0 \qquad \text{(compaction)}$$

where

$$\hat{\boldsymbol{\sigma}}(\mathbf{v}_s) = 2\mu_s(1-\phi)\Big(\mathcal{D}\mathbf{v}_s - \tfrac{1}{3}\nabla \cdot \mathbf{v}_s \mathbf{I}\Big) \qquad \text{(deviatoric stress of mixture)}$$

$$\mathcal{D}\mathbf{v}_s = \frac{1}{2}(\nabla \mathbf{v}_s + \nabla \mathbf{v}_s^T) \qquad \text{(symmetric gradient)}$$

Simple boundary conditions (for the talk)

$$\mathbf{u} \cdot \nu = 0 \quad \text{and} \quad \mathbf{v}_s = \mathbf{0} \quad \text{on } \partial\Omega$$

Advantage. A single two-phase model holds when one phase disappears. Model avoids the free boundary between one and two phase regions. Important in a time-dependent problem when the free boundary evolves.

Price. A degenerate system when the fluid melt disappears ($\phi = 0$).

1. Stokes system is well-behaved
2. Darcy system degenerates when ϕ vanishes. We will show

$$\|\phi^{-1-\Theta}\mathbf{u}\| + \|\phi^{-1/2}\nabla \cdot \mathbf{u}\| + \|\phi^{1/2} q_f\| \leq C$$

for some constant C, where $\|\cdot\|$ is the $L^2(\Omega)$-norm.

Where $\phi = 0$,

- p_f (or q_f) may be *unbounded*.
- The conservation equation is lost ($0 = 0$).
- If we impose a small nonzero porosity ϕ_0 everywhere for numerical approximation, the condition number grows as $\phi_0 \to 0$.

2. A Well-Posed Variational Problem

We use the function spaces

$$H_0(\mathrm{div};\Omega) = \left\{\mathbf{v} \in (L^2(\Omega))^d : \nabla\cdot\mathbf{v} \in L^2(\Omega),\ \mathbf{v}\cdot\nu = 0 \text{ on } \partial\Omega\right\}$$
$$(H_0^1(\Omega))^d = \left\{\mathbf{v} \in (H^1(\Omega))^d : \mathbf{v} = \mathbf{0} \text{ on } \partial\Omega\right\}$$
$$L^2(\Omega)/\mathbb{R} = \left\{w \in L^2(\Omega) : \int_\Omega w(x)\,dx = 0\right\}$$

Standard weak formulation. Assume $\phi \geq \phi_* > 0$, but consider $\phi_* \to 0$.

Find $\mathbf{u} \in H_0(\mathrm{div})$, $q_f \in L^2$, $\mathbf{v}_s \in (H_0^1)^d$, and $q \in L^2/\mathbb{R}$ such that

$$\begin{aligned}
&\left(\frac{\mu_f}{k_0}\phi^{-2-2\Theta}\mathbf{u}, \boldsymbol{\psi}_r\right) - (q_f, \nabla\cdot\boldsymbol{\psi}_r) &&= 0 &&\forall\,\boldsymbol{\psi}_r \in H_0(\mathrm{div})\\
&(\nabla\cdot\mathbf{u}, w_f) + \left(\frac{\phi}{\mu_s(1-\phi)}(q_f - q), w_f\right) &&= 0 &&\forall w_f \in L^2\\
&-(q, \nabla\cdot\boldsymbol{\psi}_s) + (\hat{\boldsymbol{\sigma}}(\mathbf{v}_s), \nabla\boldsymbol{\psi}_s) &&= (\mathbf{G}, \boldsymbol{\psi}_s) &&\forall\,\boldsymbol{\psi}_s \in (H_0^1)^d\\
&(\nabla\cdot\mathbf{v}_s, w) - \left(\frac{\phi}{\mu_s(1-\phi)}(q_f - q), w\right) &&= 0 &&\forall w \in L^2/\mathbb{R}
\end{aligned}$$

We can take the test functions

$$\begin{aligned}
\boldsymbol{\psi}_r &= \mathbf{u} \in H_0(\mathrm{div}) \quad\text{and}\quad w_f = q_f + \phi^{-1}\nabla\cdot\mathbf{u} \in L^2\\
\boldsymbol{\psi}_s &= \mathbf{v}_s \in (H_0^1)^d \quad\text{and}\quad w = q \in L^2/\mathbb{R}
\end{aligned}$$

Theorem. If $\phi \geq \phi_* > 0$ and a solution exists, then

$$\|\phi^{-1-\Theta}\mathbf{u}\| + \|\phi^{-1/2}\nabla\cdot\mathbf{u}\| + \|\phi^{1/2}q_f\| + \|\mathbf{v}_s\|_1 + \|q\| \leq C\|\phi^{1/2}q_s\|$$

Problems. If $\phi = 0$ (where there is no fluid!):

- We have no control on the size of the fluid potential q_f, so $q_f \notin L^2$
- We lose the conservation equation $\nabla\cdot(\phi\mathbf{v}_r) + \phi q_f = \phi q_s$.
- Some of the integrals are ill defined as $\phi \to 0$ (division by ϕ).

Key idea. Scale the variables so these energy bounds make sense.

Define a *scaled potential and velocity*

$$\boxed{\tilde{q}_f = \phi^{1/2}q_f \quad \text{and} \quad \tilde{\mathbf{v}}_r = \phi^{-1-\Theta}\mathbf{u} = \phi^{-\Theta}\mathbf{v}_r}$$

Darcy's law for the fluid

$$\tilde{\mathbf{v}}_r + \frac{k_0 \phi^{1+\ominus}}{\mu_f} \nabla(\phi^{-1/2} \tilde{q}_f) = 0 \quad \text{(Darcy law)}$$

$$\mu_s \phi^{-1/2} \nabla \cdot (\phi^{1+\ominus} \tilde{\mathbf{v}}_r) + \frac{1}{1-\phi}\left(\tilde{q}_f - \phi^{1/2} q\right) = 0 \quad \text{(conservation/ compaction)}$$

Stokes system for a highly viscous, compressible matrix plus fluid

$$\nabla q - \nabla \cdot \hat{\boldsymbol{\sigma}}(\mathbf{v}_s) = \mathbf{G} \quad \text{(Stokes)}$$

$$\mu_s \nabla \cdot \mathbf{v}_s - \frac{\phi^{1/2}}{1-\phi}\left(\tilde{q}_f - \phi^{1/2} q\right) = 0 \quad \text{(compaction)}$$

Simple boundary conditions

$$\phi^{1+\ominus} \tilde{\mathbf{v}}_r \cdot \nu = 0 \quad \text{and} \quad \mathbf{v}_s = \mathbf{0} \quad \text{on } \partial\Omega$$

Remarks. Where $\phi = 0$:

- The scaled fluid pressure $\tilde{q}_f$ remains defined and bounded
- We do not lose the second equation

Questions.

- Does $\phi^{-1/2} \nabla \cdot (\phi^{1+\ominus} \tilde{\mathbf{v}}_r)$ make mathematical sense?
- What Hilbert space does $\tilde{\mathbf{v}}_r$ lie within?

Formally,

$$\phi^{-1/2}\nabla\cdot(\phi^{1+\ominus}\mathbf{v}) = (1+\ominus)\phi^{\ominus-1/2}\nabla\phi\cdot\mathbf{v} + \phi^{1/2+\ominus}\nabla\cdot\mathbf{v}$$

Assumption. $\boxed{\phi \in L^\infty(\Omega) \text{ and } \phi^{\ominus-1/2}\nabla\phi \in (L^\infty(\Omega))^d}$

Define:

$$H_\phi(\text{div};\Omega) = \left\{\mathbf{v}\in(L^2(\Omega))^d \,:\, \phi^{-1/2}\nabla\cdot(\phi^{1+\ominus}\mathbf{v}) \in L^2(\Omega)\right\}$$

That is, $\phi^{1/2+\ominus}$ times the weak divergence of $\mathbf{v}$ lies in L^2.

Lemma. $H_\phi(\text{div};\Omega)$ is a Hilbert space with the inner-product

$$(\mathbf{u},\mathbf{v})_{H_\phi(\text{div})} = (\mathbf{u},\mathbf{v}) + \left(\phi^{-1/2}\nabla\cdot(\phi^{1+\ominus}\mathbf{u}), \phi^{-1/2}\nabla\cdot(\phi^{1+\ominus}\mathbf{v})\right).$$

Moreover, $H(\text{div};\Omega) = H_1(\text{div};\Omega) \subset H_\phi(\text{div};\Omega)$.

We wish to apply the homogeneous Neumann boundary condition.

Define: $\gamma_\phi : H_\phi(\mathrm{div};\Omega) \to H^{-1/2}(\partial\Omega)$ by the integration by parts formula

$$\langle \gamma_\phi(\mathbf{v}), w\rangle = \left(\phi^{-1/2}\nabla\cdot(\phi^{1+\Theta}\mathbf{v}), w\right) + \left(\mathbf{v}, \phi^{1+\Theta}\nabla(\phi^{-1/2}w)\right),$$

wherein $w \in H^{1/2}(\partial\Omega)$ has been extended to $w \in H^1(\Omega)$.

We should interpret $\gamma_\phi(\mathbf{v}) = \phi^{1/2+\Theta}\mathbf{v}\cdot\nu$.

Lemma. The trace operator

$$\gamma_\phi : H_\phi(\mathrm{div};\Omega) \to H^{-1/2}(\partial\Omega)$$

is well defined and

$$\|\phi^{1/2+\Theta}\mathbf{v}\cdot\nu\|_{-1/2,\partial\Omega} \le C\|\mathbf{v}\|_{H_\phi(\mathrm{div})} \quad \forall \mathbf{v} \in H_\phi(\mathrm{div};\Omega)$$

Define:

$$H_{\phi,0}(\mathrm{div};\Omega) = \left\{\mathbf{v} \in H_\phi(\mathrm{div};\Omega) \;:\; \phi^{1/2+\Theta}\mathbf{v}\cdot\nu = 0 \text{ on } \partial\Omega\right\}$$

$$H_{\phi,0}(\text{div};\Omega) = \Big\{\mathbf{v} \in (L^2(\Omega))^d : \phi^{-1/2}\nabla\cdot(\phi^{1+\Theta}\mathbf{v}) \in L^2(\Omega),\ \phi^{1/2+\Theta}\mathbf{v}\cdot\nu = 0 \text{ on } \partial\Omega\Big\}$$

Scaled weak formulation.

Find $\tilde{\mathbf{v}}_r \in H_{\phi,0}(\text{div})$, $\tilde{q}_f \in L^2$, $\mathbf{v}_s \in (H_0^1)^d$, and $q \in L^2/\mathbb{R}$ such that

$$\Big(\frac{\mu_f}{k_0}\tilde{\mathbf{v}}_r, \boldsymbol{\psi}_r\Big) - \Big(\tilde{q}_f, \phi^{-1/2}\nabla\cdot(\phi^{1+\Theta}\boldsymbol{\psi}_r)\Big) = 0 \ \forall \boldsymbol{\psi}_r \in H_{\phi,0}(\text{div})$$

$$\Big(\phi^{-1/2}\nabla\cdot(\phi^{1+\Theta}\tilde{\mathbf{v}}_r), w_f\Big) + \Big(\frac{1}{\mu_s(1-\phi)}(\tilde{q}_f - \phi^{1/2}q), w_f\Big) = 0 \ \forall w_f \in L^2$$

$$-(q, \nabla\cdot\boldsymbol{\psi}_s) + (\hat{\boldsymbol{\sigma}}(\mathbf{v}_s), \nabla\boldsymbol{\psi}_s) = (\mathbf{G}, \boldsymbol{\psi}_s) \ \forall \boldsymbol{\psi}_s \in (H_0^1)^d$$

$$(\nabla\cdot\mathbf{v}_s, w) - \Big(\frac{\phi^{1/2}}{\mu_s(1-\phi)}(\tilde{q}_f - \phi^{1/2}q), w\Big) = 0 \ \forall w \in L^2/\mathbb{R}$$

Theorem. There exists a unique solution and

$$\|\tilde{\mathbf{v}}_r\| + \|\phi^{-1/2}\nabla\cdot(\phi^{1+\Theta}\tilde{\mathbf{v}}_r)\| + \|\tilde{q}_f\| + \|\mathbf{v}_s\|_1 + \|q\| \le C|\rho_r|.$$

> The scaled model is well-posed even if the fluid phase disappears

Idea of proof.

The proof requires estimates for Darcy and Stokes problems.
The Babuška-Lax-Milgram Theorem gives existence and uniqueness (after adding divergence stabilization terms to the bilinear form).

3. A Mixed Finite Element Method

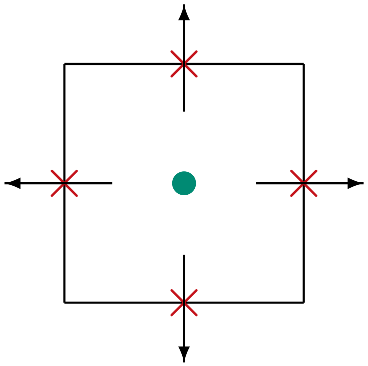
$\Omega \subset \mathbb{R}^d$ is a polygonal domain. Assume $d = 2$ dimensions for the talk.

RT$_0$ *for Darcy.* $\mathbb{V}_{\mathsf{RT}} \times \mathbb{W}_{\mathsf{RT}} \subset H_0(\mathrm{div};\Omega) \times L^2(\Omega)$.

Lowest order Raviart-Thomas finite element space

On an element $E \in \mathcal{T}_h$, $(\mathbb{P}_{1,0} \times \mathbb{P}_{0,1}) \times \mathbb{P}_0$

Accuracy $\mathcal{O}(h)$ for $\tilde{\mathbf{v}}_r$ and $\tilde{q}_f$.

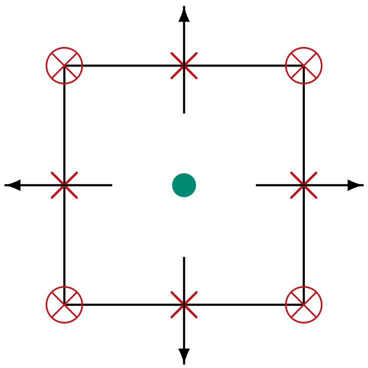


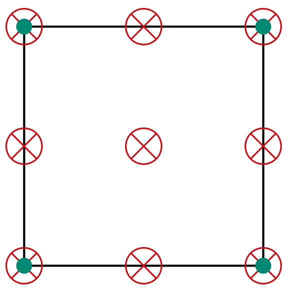
Any Stokes elements. $\mathbb{V}_{\mathsf{S}} \times \mathbb{W}_{\mathsf{S}} \subset (H_0^1(\Omega))^2 \times L^2(\Omega)$

BR: Bernardi-Raugel.

On $E \in \mathcal{T}_h$, $(\mathbb{P}_{1,2} \times \mathbb{P}_{2,1}) \times \mathbb{P}_0$

Accuracy $\mathcal{O}(h)$ for $\mathbf{v}_s$ (in H^1) and q_s.





TH: Taylor-Hood (lowest order)

On $E \in \mathcal{T}_h$, $\mathbb{P}^2_{2,2} \times \mathbb{P}_{1,1}$

Accuracy $\mathcal{O}(h^2)$ for $\mathbf{v}_s$ (in H^1) and q_s.

Scaled mixed finite element method.

Find $\tilde{\mathbf{v}}_{r,h} \in \mathbb{V}_{\mathsf{RT}}$, $\tilde{q}_{f,h} \in \mathbb{W}_{\mathsf{RT}}$, $\mathbf{v}_s \in \mathbb{V}_{\mathsf{S}}$, and $q_h \in \mathbb{W}_{\mathsf{S}}$ such that

$$\begin{aligned}
&\Big(\frac{\mu_f}{k_0}\tilde{\mathbf{v}}_{r,h}, \boldsymbol{\psi}_r\Big) - \Big(\tilde{q}_{f,h}, \phi^{-1/2}\nabla\cdot(\phi^{1+\Theta}\boldsymbol{\psi}_r)\Big) && = 0 \ \forall \boldsymbol{\psi}_r \in \mathbb{V}_{\mathsf{RT}} \\
&\Big(\phi^{-1/2}\nabla\cdot(\phi^{1+\Theta}\tilde{\mathbf{v}}_{r,h}), w_f\Big) + \Big(\frac{1}{\mu_s(1-\phi)}(\tilde{q}_{f,h} - \phi^{1/2}q_h), w_f\Big) && = 0 \ \forall w_f \in \mathbb{W}_{\mathsf{RT}} \\
&-(q_h, \nabla\cdot\boldsymbol{\psi}_s) + (\hat{\boldsymbol{\sigma}}(\mathbf{v}_{s,h}), \nabla\boldsymbol{\psi}_s) && = (\mathbf{G}, \boldsymbol{\psi}_s) \ \forall \boldsymbol{\psi}_s \in \mathbb{V}_{\mathsf{S}} \\
&(\nabla\cdot\mathbf{v}_{s,h}, w) - \Big(\frac{\phi^{1/2}}{\mu_s(1-\phi)}(\tilde{q}_{f,h} - \phi^{1/2}q_h), w\Big) && = 0 \ \forall w \in \mathbb{W}_{\mathsf{S}}
\end{aligned}$$

Lemma. There exists a unique solution to the discrete equations.

Concern: How do we handle $\phi^{-1/2}\nabla\cdot(\phi^{1+\Theta}\mathbf{v})$ in a code?

It scales as $\phi^{1/2+\Theta}$, so set it to zero when $\phi = 0$ (at a quadrature point).

Define the discrete Darcy velocity $\mathbf{u}_h \in \mathbb{V}_{\mathsf{RT}}$ and fluid potential $q_{f,h} \in \mathbb{W}_{\mathsf{RT}}$ by their degrees of freedom:

$$\mathbf{u} = \phi^{1+\Theta}\tilde{\mathbf{v}}_r \quad \Longrightarrow \quad \mathbf{u}_h \cdot \nu|_e = \frac{1}{|e|}\int_e \phi^{1+\Theta}\,ds\;\tilde{\mathbf{v}}_{r,h}\cdot\nu|_e \qquad \forall e \in \mathcal{E}_h$$

$$q_f = \phi^{-1/2}\tilde{q}_f \quad \Longrightarrow \quad q_{f,h}|_E = \phi_E^{-1/2}\tilde{q}_{f,h}|_E \qquad \forall E \in \mathcal{T}_h$$

We arbitrarily set $q_{f,h}|_E = 0$ if $\phi_E = \dfrac{1}{|E|}\displaystyle\int \phi\,dx = 0$.

RT-projections.

- $\mathcal{P}_{\text{RT}} = \hat{\cdot} : L^2(\Omega) \to \mathbb{W}_{\text{RT}}$ is $L^2(\Omega)$-projection onto piecewise constants
- $\pi_{\text{RT}} : H(\text{div};\Omega) \cap L^{2+\epsilon}(\Omega) \to \mathbb{V}_{\text{RT}}$ (any $\epsilon > 0$) is the standard Raviart-Thomas or Fortin operator

S-projections.

- $\mathcal{P}_{\text{S}} : L^2(\Omega) \to \mathbb{W}_{\text{S}}$ is $L^2(\Omega)$-projection onto $\mathbb{W}_{\text{S}}$
- $\pi_{\text{S}} : H^1(\Omega) \to \mathbb{V}_{\text{S}}$ is $H^1(\Omega)$-projection onto $\mathbb{V}_{\text{S}}$

Theorem. If $\mathcal{T}_h$ is quasi-uniform,

$$
\begin{aligned}
\|\tilde{\mathbf{v}}_r - \tilde{\mathbf{v}}_{r,h}\| &+ \left\|\mathcal{P}_{\text{RT}}\left[\phi^{-1/2}\nabla\cdot(\phi^{1+\Theta}(\tilde{\mathbf{v}}_r - \tilde{\mathbf{v}}_{r,h}))\right]\right\| + \|\mathbf{v}_s - \mathbf{v}_{s,h}\|_1 \\
&+ \|\tilde{q}_f - \tilde{q}_{f,h}\| + \|q - q_h\| \\
&\le C\Big\{\|\tilde{\mathbf{v}}_r - \pi_{\text{RT}}\tilde{\mathbf{v}}_r\| + \left\|\mathcal{P}_{\text{RT}}\left[\phi^{-1/2}\nabla\cdot(\phi^{1+\Theta}(\tilde{\mathbf{v}}_r - \pi_{\text{RT}}\tilde{\mathbf{v}}_r))\right]\right\| \\
&\qquad + \|q - \mathcal{P}_{\text{S}}q\| + \|\mathbf{v}_s - \pi_{\text{S}}\mathbf{v}_s\|_1 + \|\tilde{q}_f - \mathcal{P}_{\text{RT}}\tilde{q}_f\|\Big\}
\end{aligned}
$$

$\boxed{\text{If the solution is sufficiently regular, the approximation is } \mathcal{O}(h).}$

A Modification for Local Mass Conservation

The mixed finite element method is not locally mass conservative.

Locally average ϕ. Let $\widehat{\phi} = \mathcal{P}_{\mathbb{W}_{\mathsf{RT}}}\phi \in \mathbb{W}_{\mathsf{RT}}$, so

$$\text{For } \mathbf{x} \in E, \quad \widehat{\phi}(\mathbf{x}) = \phi_E = \frac{1}{|E|}\int_E \phi\, dx$$

When $\widehat{\phi}|_E = \phi_E = 0$, ϕ is identically zero on E.

Modification. Replace

$$\phi^{-1/2}\nabla\cdot(\phi^{1+\ominus}\mathbf{v})\Big|_E \quad \text{by} \quad \begin{cases} \phi_E^{-1/2}\nabla\cdot(\phi^{1+\ominus}\mathbf{v}) & \text{if } \phi_E \neq 0, \\ 0 & \text{if } \phi_E = 0 \end{cases}$$

and modify two terms involving the pressure potentials.

Locally conservative scaled mixed finite element method. Find $\tilde{\mathbf{v}}_{r,h} \in \mathbb{V}_{\mathsf{RT}}$, $\tilde{q}_{f,h} \in \mathbb{W}_{\mathsf{RT}}$, $\mathbf{v}_{s,h} \in \mathbb{V}_{\mathsf{S}}$, and $q_h \in \mathbb{W}_{\mathsf{S}}$ such that

$$\left(\frac{\mu_f}{k_0}\tilde{\mathbf{v}}_{r,h}, \boldsymbol{\psi}_r\right) - \left(\tilde{q}_{f,h}, \widehat{\phi}^{-1/2}\nabla\cdot(\phi^{1+\Theta}\boldsymbol{\psi}_r)\right) = 0 \ \forall \boldsymbol{\psi}_r \in \mathbb{V}_{\mathsf{RT}}$$

$$\left(\widehat{\phi}^{-1/2}\nabla\cdot(\phi^{1+\Theta}\tilde{\mathbf{v}}_{r,h}), w_f\right) + \left(\frac{\phi\widehat{\phi}^{-1}}{\mu_s(1-\phi)}(\tilde{q}_{f,h} - \widehat{\phi}^{1/2}q_h), w_f\right) = 0 \ \forall w_f \in \mathbb{W}_{\mathsf{RT}}$$

$$-(q_h, \nabla\cdot\boldsymbol{\psi}_s) + (\widehat{\boldsymbol{\sigma}}(\mathbf{v}_{s,h}), \nabla\boldsymbol{\psi}_s) = (\mathbf{G}, \boldsymbol{\psi}_s) \ \forall \boldsymbol{\psi}_s \in \mathbb{V}_{\mathsf{S}}$$

$$(\nabla\cdot\mathbf{v}_{s,h}, w) - \left(\frac{\phi\widehat{\phi}^{-1/2}}{\mu_s(1-\phi)}(\tilde{q}_{f,h} - \widehat{\phi}^{1/2}q_h), w\right) = 0 \ \forall w \in \mathbb{W}_{\mathsf{S}}$$

Where $\widehat{\phi}|_E = \phi_E = 0$, set $\widehat{\phi}^{-1/2} = 0$ and interpret $\phi\widehat{\phi}^{-1} = 1$.

Local conservation. Take

$$w_f = \widehat{\phi}^{1/2} \in \mathbb{W}_{\mathsf{RT}} \qquad \text{and} \qquad w = 1 \in \mathbb{W}_{\mathsf{S}}$$

$\Longrightarrow$

$$0 = \int_E \left[\nabla\cdot(\phi^{1+\Theta}\tilde{\mathbf{v}}_{r,h}) + \nabla\cdot\mathbf{v}_{s,h}\right]dx = \int_E \left[\nabla\cdot\mathbf{u}_h + \nabla\cdot\mathbf{v}_{s,h}\right]dx$$

For the locally conservative method, we can show

- The existence of a unique discrete solution
- The method is stable.

We cannot show

- convergence of the locally conservative method at this time

However

- the numerical results show optimal convergence and often superconvergence

Implementation

$$\begin{pmatrix} A & -B_\phi & 0 & 0 \\ B_\phi^T & C_{f,\phi} & 0 & -C_{f,s,\phi} \\ 0 & 0 & D_\phi & -B \\ 0 & -C_{f,s,\phi}^T & B^T & C_{s,\phi} \end{pmatrix} \begin{pmatrix} \tilde{v}_r \\ \tilde{q}_f \\ v_s \\ q \end{pmatrix} = \begin{pmatrix} a_\phi \\ 0 \\ b_\phi \\ 0 \end{pmatrix},$$

Evaluation of B_ϕ. Use the divergence theorem.
For the locally conservative method and element $E \in \mathcal{T}_h$ and edge $e \in \mathcal{E}_h$,

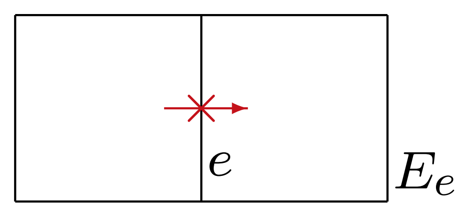
$$\begin{aligned} B_{\phi,e,E} &= \left(\hat{\phi}^{-1/2}\nabla \cdot (\phi^{1+\Theta}\mathbf{v}_e), w_E\right) \\ &= \begin{cases} \phi_E^{-1/2}\displaystyle\int_e \phi^{1+\Theta}\, ds\, \mathbf{v}_e \cdot \nu_E & \text{if } e \subset \partial E \text{ and } \phi_E \neq 0, \\ 0 & \text{if } e \not\subset \partial E \text{ or } \phi_E = 0. \end{cases} \end{aligned}$$

Similarly for the nonconservative scaled method.

Mass Lumping on Rectangular Meshes. To eliminate $\tilde{v}_r$, diagonalize A using reduced quadrature (mass lumping). (Russell & Wheeler, 1983)

For edges $e, f \in \mathcal{E}_h$, apply rectangle rule

$$A_{e,f} = \left(\frac{\mu_f}{k_0}\mathbf{v}_e, \mathbf{v}_f\right)_Q \approx \left(\frac{\mu_f}{k_0}\mathbf{v}_e, \mathbf{v}_f\right)_Q = \frac{\mu_f}{2k_0}|E_e|\delta_{e,f}$$



What remains is a Stokes-like system with *two* pressure potentials.

4. Some Numerical Results

Numerical Results for a Simple Degenerate Darcy Problem

The Stokes part is well-posed, since $\phi \le \phi^* < 1$ (rock always present).

Simple degenerate Darcy problem. Recall $\mathbf{u} = \phi \mathbf{v}_r$.

$$
\begin{aligned}
\mathbf{v}_r &= -\phi^{1+2\Theta} \nabla q_f && \text{in } \Omega && \text{(Darcy law)} \\
\nabla \cdot (\phi \mathbf{v}_r) + \phi q_f &= \phi q_s && \text{in } \Omega && \text{(conservation)} \\
\phi^{1/2+\Theta} \mathbf{v}_r \cdot \nu &= 0 && \text{on } \partial\Omega && \text{(no flow BC)}
\end{aligned}
$$

Scaled mixed finite element method.

$$
\boxed{\tilde{q}_f = \phi^{1/2} q_f \quad \text{and} \quad \tilde{\mathbf{v}}_r = \phi^{-\Theta} \mathbf{v}_r}
$$

Find $\tilde{\mathbf{v}}_{r,h} \in \mathbb{V}_{\mathrm{RT}}$, $\tilde{q}_{f,h} \in \mathbb{W}_{\mathrm{RT}}$, $\mathbf{v}_s \in \mathbb{V}_{\mathrm{S}}$, and $q_h \in \mathbb{W}_{\mathrm{S}}$ such that

$$
\begin{aligned}
(\tilde{\mathbf{v}}_{r,h}, \boldsymbol{\psi}_r) - \left(\tilde{q}_{f,h}, \phi^{-1/2} \nabla \cdot (\phi^{1+\Theta} \boldsymbol{\psi}_r)\right) &= 0 \ \forall \boldsymbol{\psi}_r \in \mathbb{V}_{\mathrm{RT}} \\
\left(\phi^{-1/2} \nabla \cdot (\phi^{1+\Theta} \tilde{\mathbf{v}}_{r,h}), w_f\right) + (\tilde{q}_{f,h} - \phi^{1/2} q_h, w_f) &= 0 \ \forall w_f \in \mathbb{W}_{\mathrm{RT}}
\end{aligned}
$$

Common features

- $\Omega = (-1,1)^d$, $d = 1,2$
- Uniform rectangular mesh of $n = 1/h$ elements in each direction
- Use the locally conservative CCFD method
- $\Theta = 0$ (so $\tilde{\mathbf{v}}_r = \mathbf{v}_r$ and $\mathbf{u} = \phi\mathbf{v}_r$)

- Closed form solutions or manufactured solutions
  - Give ϕ and q_f
  - Compute the RHS $f = \phi^{1/2} q_s$ and BC terms

- Use discrete L^2-norms to measure relative errors
  - Midpoint quadrature $\longleftrightarrow$ $\|\widehat{\tilde{q}}_f - \tilde{q}_{f,h}\|$ and $\|\widehat{q}_f - q_{f,h}\|$
  - Trapezoidal quadrature $\longleftrightarrow$ $\|\pi_{\mathrm{RT}}\mathbf{v}_r - \mathbf{v}_{r,h}\|$

  These are norms for which superconvergence might be expected.

Let $\Omega = (-1,1)$ and

$$\phi(x) = \begin{cases} 0, & x < 0 \\ x^2, & x > 0 \end{cases} \quad \text{and} \quad f(x) = x^{1+\beta}, \quad 0 < x < 1$$

Then we have the Euler equation

$$\boxed{-x^2\, q_f'' - 4x\, q_f' + q_f = x^\beta, \quad 0 < x < 1}$$

The Euler exponents are

$$r_1 = \frac{-3+\sqrt{13}}{2} \approx 0.3 > 0 \quad \text{and} \quad r_2 = \frac{-3-\sqrt{13}}{2} \approx -3.3 < 0$$

Since $u = -\phi q_f' \in L^2(0,1)$, x^{r_2} is not part of the solution.

If $\beta \neq r_1, r_2$, we have the closed form solution

$$q_f(x) = \begin{cases} 0, & -1 < x \le 0 \\ \dfrac{\beta x^{r_1} - r_1 x^\beta}{r_1(\beta - r_1)(\beta - r_2)}, & 0 < x < 1 \end{cases} \quad \text{and} \quad \tilde{q}_f(x) = x\, q_f(x)$$

$$u(x) = \begin{cases} 0, & -1 < x \le 0 \\ \dfrac{-\beta(x^{r_1+1} - x^{\beta+1})}{(\beta - r_1)(\beta - r_2)}, & 0 < x < 1 \end{cases}$$

Regularity. The potential singularity near $x = 0$ is

$$\tilde{q}_f \sim \mathbf{v}_r \sim |x|^{1.3} + |x|^{1+\beta} \quad \text{and} \quad q_f \sim |x|^{0.3} + |x|^{\beta}$$

Thus

$$\tilde{q}_f, u \in H^{\min(1.8, 1.5+\beta)} \quad \text{and} \quad q_f \in H^{\min(0.8, 0.5+\beta)}$$

Condition on ϕ. Since $\phi(x) = x^2$,

$$0 < x < 1 \quad \Longrightarrow \quad \phi^{-1/2}\phi' = 2 \in L^\infty(0,1)$$

Euler Test — 2, Relative discrete L^2-norm errors

β	n	scaled pressure $\tilde{q}_f$ error	rate	pressure q_f err	rate	velocity u err	rate
0.5	64	0.000642	1.669	0.004341	0.638	0.002387	1.640
	128	0.000199	1.691	0.002724	0.672	0.000754	1.663
	256	0.000061	1.709	0.001681	0.697	0.000235	1.681
	512	0.000018	1.723	0.001024	0.716	0.000073	1.695
			1.8		0.8		1.8
−0.5	64	0.000802	1.254	0.039971	0.013	0.006749	0.976
	128	0.000358	1.166	0.039289	0.025	0.003426	0.978
	256	0.000167	1.101	0.038474	0.030	0.001731	0.985
	512	0.000080	1.060	0.037617	0.033	0.000872	0.990
			1		0		1
−1.0	64	0.004849	0.396	0.164987	-0.089	0.010550	0.546
	128	0.003526	0.460	0.170768	-0.050	0.007338	0.524
	256	0.002521	0.484	0.173955	-0.027	0.005142	0.513
	512	0.001790	0.494	0.175645	-0.014	0.003618	0.507
			0.5		???		0.5
−1.5	64	0.059596	0.016	0.278083	-0.023	0.003470	0.473
	128	0.058620	0.024	0.279416	-0.007	0.002856	0.281
	256	0.057593	0.026	0.279819	-0.002	0.002507	0.188
	512	0.056590	0.025	0.279939	-0.001	0.002281	0.137
			0		???		0

Manufactured solution. We assume that

$$\phi = \begin{cases} 0 & x \leq -3/4 \text{ or } y \leq -3/4, \\ (x+3/4)^{\alpha}(y+3/4)^{2\alpha} & \text{otherwise,} \end{cases}$$
$$q_f = \cos(6xy^2)$$

Smoothness of ϕ. Note that the singularity is

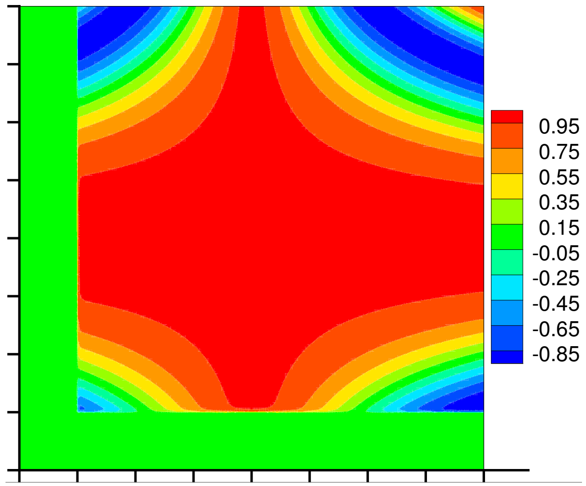
$$\phi^{-1/2}\nabla\phi \sim (x+3/4)^{\alpha/2-1}$$

Thus

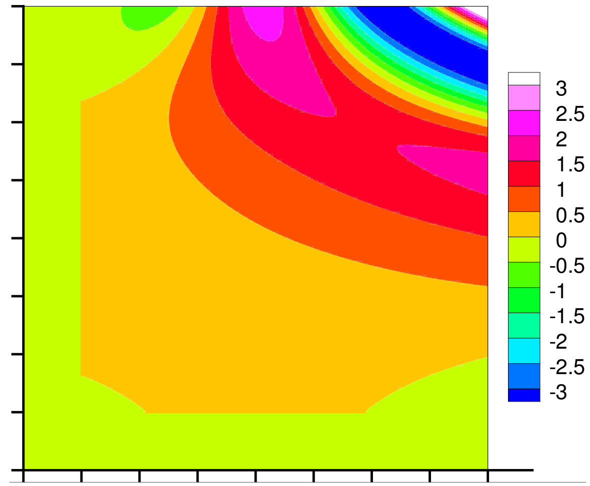
$$\phi^{-1/2}\nabla\phi \in L^{\infty}((-1,1)^2) \quad \text{only if} \quad \alpha \geq 2$$

Two-Dimensional Test — 2

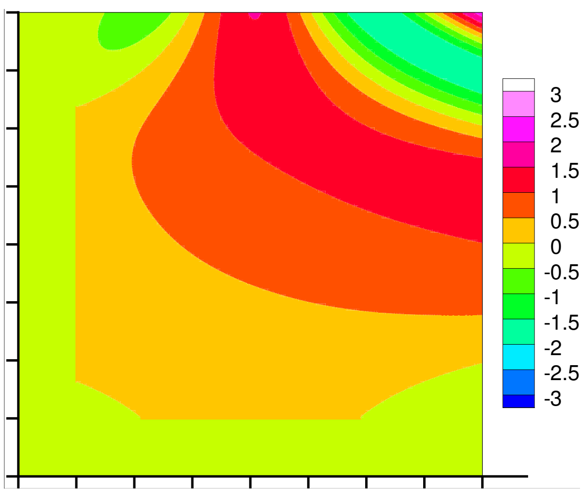
The solution. Problems when a phase is lost ($\phi = 0$).



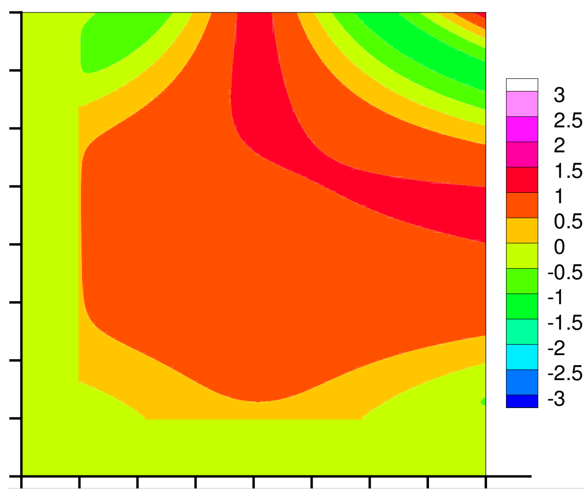
Pressure q_f



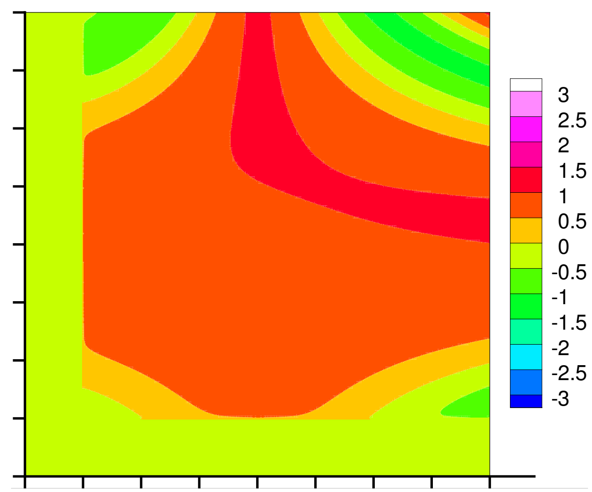
$\tilde{q}_f$, $\alpha = 2$



$\tilde{q}_f$, $\alpha = 1$



$\tilde{q}_f$, $\alpha = 1/4$



$\tilde{q}_f$, $\alpha = 1/8$

Relative discrete L^2-norm errors,
missing the one to two phase boundary

α	n	scaled pressure $\tilde{q}_f$ error	scaled pressure $\tilde{q}_f$ rate	pressure q_f error	pressure q_f rate	velocity u error	velocity u rate
2	33	0.012137	—	0.021447	—	0.028001	—
	65	0.003171	1.980	0.007832	1.486	0.009146	1.651
	129	0.000817	1.979	0.002769	1.517	0.002753	1.752
	257	0.000210	1.971	0.000969	1.523	0.000790	1.811
	513	0.000055	1.938	0.000339	1.520	0.000220	1.850
			2		1.5		2
0.25	33	0.031315	—	0.047229	—	0.039155	—
	65	0.020907	0.596	0.029933	0.673	0.024967	0.664
	129	0.014105	0.574	0.019492	0.626	0.017147	0.548
	257	0.009588	0.560	0.012969	0.591	0.012062	0.510
	513	0.006566	0.548	0.008778	0.565	0.008545	0.499
			0.5		0.5		0.5

Numerical Results for the Full Model

A mantle column for $z \in [-L, L]$

Boundary conditions: No flow and set potential scale

$$v_s(-L) = v_s(L) = u(-L) = u(L) = 0 \quad \text{and} \quad q_f(0) = 0$$

Non-dimensionalized equations: Where $\phi > 0$, for $u = \phi v_r$

$$\phi^{2+2\Theta}\left(\frac{3 + \phi - 4\phi^2}{3\phi} u'\right)' - u = \phi^{2+2\Theta}(1 - \phi)$$

$$v_s = -u' \qquad q_s = \frac{1}{\phi} u' + q_f \qquad q_f' = -\frac{1}{\phi^{2+2\Theta}} u$$

Where $\phi = 0$,

$$u = 0 \qquad q_s' = 1 \qquad v_s' = 0$$

$$\phi(z) = \begin{cases} \phi_- & \text{if } z \le 0 \\ \phi_+ & \text{if } z > 0 \end{cases}$$

Solution.

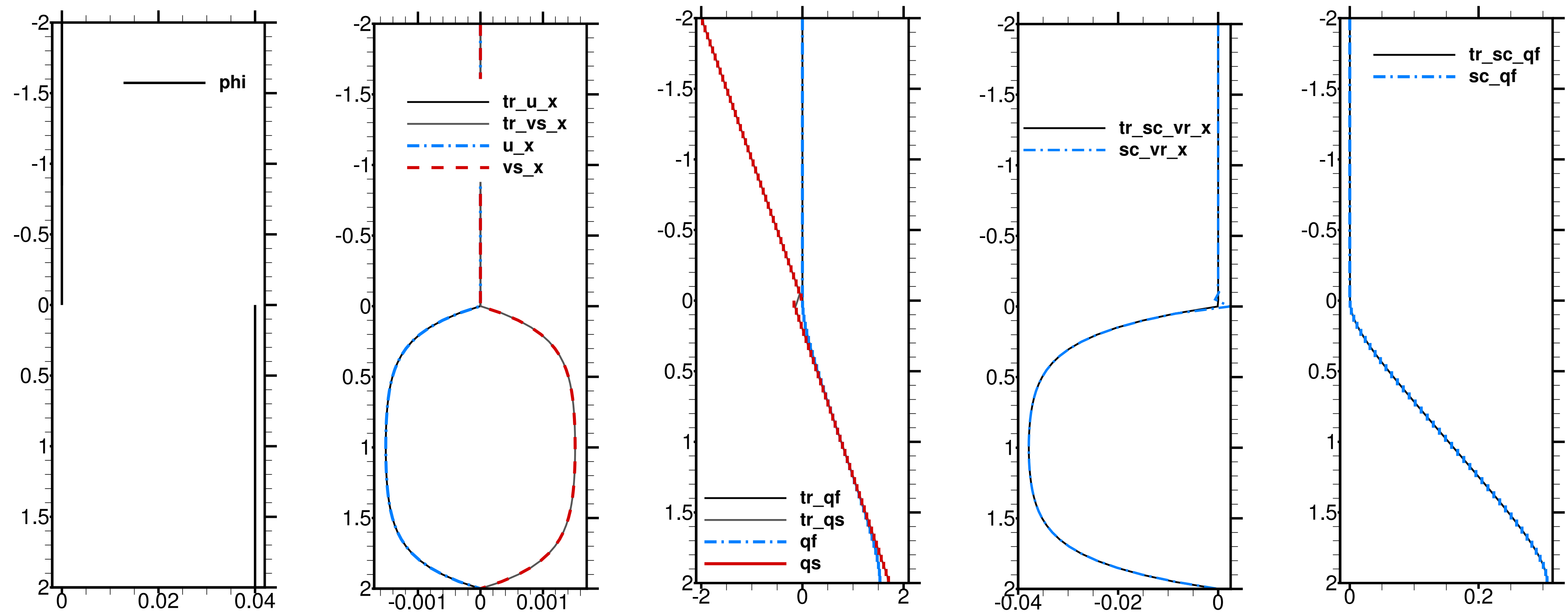
$$u_\pm = \phi_\pm^{2+2\Theta}(1-\phi_\pm)\Big[1 + a_\pm \cosh(R_\pm z) + b_\pm \sinh(R_\pm z)\Big]$$

- If both $\phi_\pm > 0$, let $\mathcal{F}_\pm = \phi_\pm^{2+2\Theta}(1-\phi_\pm)$ and solve

$$\begin{bmatrix} \cosh(R_+L) & 0 & \sinh(R_+L) & 0 \\ 0 & \cosh(R_-L) & 0 & -\sinh(R_-L) \\ \mathcal{F}_+ & -\mathcal{F}_- & 0 & 0 \\ 0 & 0 & R_-(1-\phi_+) & -R_+(1-\phi_-) \end{bmatrix} \begin{bmatrix} a_+ \\ a_- \\ b_+ \\ b_- \end{bmatrix} = \begin{bmatrix} -1 \\ -1 \\ \mathcal{F}_- - \mathcal{F}_+ \\ 0 \end{bmatrix}$$

- If $\phi_- = 0$, $a_- = b_- = 0$ $(u_- = v_{m-} = 0)$

$$a_+ = -1 \qquad b_+ = \frac{\cosh(R_+L) - 1}{\sinh(R_+L)} \qquad q_{m-} = x - b_+\frac{1-\phi_+}{R_+}$$



Porosity ϕ | $u,\ v_s$ | $q_f,\ q_s$ | Scaled $\tilde{v}_r$ | Scaled $\tilde{q}_f$

Relative L^2 potential errors and convergence rates

$(h = 4/n,\ L = 2,\ \Theta = 0,\ \phi_- = 0,\ \phi_+ = 0.04)$

n	$\tilde{q}_f$ error	$\tilde{q}_f$ rate	q error	q rate	Midpoint $\tilde{q}_f$ error	Midpoint $\tilde{q}_f$ rate	Midpoint q error	Midpoint q rate
10	2.08e-2	—	7.25e-2	—	2.98e-3	—	3.38e-4	—
20	1.04e-2	1.00	3.62e-2	1.00	9.54e-4	1.64	1.23e-4	1.46
40	5.20e-3	1.00	1.81e-2	1.00	2.53e-4	1.91	3.86e-5	1.67
80	2.60e-3	1.00	9.06e-3	1.00	6.22e-5	2.02	1.10e-5	1.81
160	1.30e-3	1.00	4.53e-3	1.00	1.50e-5	2.05	2.96e-6	1.89
		1		1		2		2

n	$\tilde{q}_f$ error	$\tilde{q}_f$ rate	q error	q rate	Interior $\tilde{q}_f$ error	Interior $\tilde{q}_f$ rate	Interior q error	Interior q rate
11	1.88e-2	—	6.94e-2	—	1.42e-2	—	4.87e-2	—
21	9.96e-3	0.98	3.95e-2	0.87	9.13e-3	0.68	3.02e-2	0.74
41	5.14e-3	0.99	2.32e-2	0.80	4.98e-3	0.90	1.66e-2	0.89
81	2.61e-3	0.99	1.43e-2	0.71	2.57e-3	0.97	8.67e-3	0.95
161	1.32e-3	1.00	9.22e-3	0.64	1.30e-3	0.99	4.43e-3	0.98
		1		0.5?		1		1

Relative L^2 velocity errors and convergence rates

$(h = 4/n,\ L = 2,\ \Theta = 0,\ \phi_- = 0,\ \phi_+ = 0.04)$

	$\tilde{v}_r$		u		v_s		Mass lumped $\tilde{v}_r$	
n	error	rate	error	rate	error	rate	error	rate
10	7.36e-3	—	1.70e-4	—	1.70e-4	—	5.61e-3	—
20	2.93e-3	1.33	4.71e-5	1.85	4.71e-5	1.85	1.69e-3	1.73
40	1.12e-3	1.39	1.21e-5	1.96	1.21e-5	1.96	4.50e-4	1.91
80	4.12e-4	1.44	3.09e-6	1.97	3.09e-6	1.97	1.14e-4	1.98
160	1.49e-4	1.47	7.85e-7	1.98	7.85e-7	1.98	2.87e-5	1.99
		1.5		2		2		2

11	4.42e-3	—	2.03e-4	—	2.03e-4	—
21	2.03e-3	1.20	9.00e-5	1.25	9.00e-5	1.25
41	1.06e-3	0.98	4.52e-5	1.03	4.52e-5	1.03
81	5.61e-4	0.93	2.37e-5	0.95	2.37e-5	0.95
161	2.93e-4	0.95	1.23e-5	0.96	1.23e-5	0.96
		1		1		1

$$\phi(z) = \begin{cases} 0 & \text{if } z \le 0, \\ \phi_+ z^2 & \text{if } z > 0, \end{cases}$$

Solution. Set $\Theta = 0$ and $\phi_+ \ll 1$. Approximately reduce to an Euler equation.

$$r_1 = \frac{3 + \sqrt{9 + 4/\phi_+}}{2} > 3 \qquad r_2 = \frac{3 - \sqrt{9 + 4/\phi_+}}{2} < 0$$

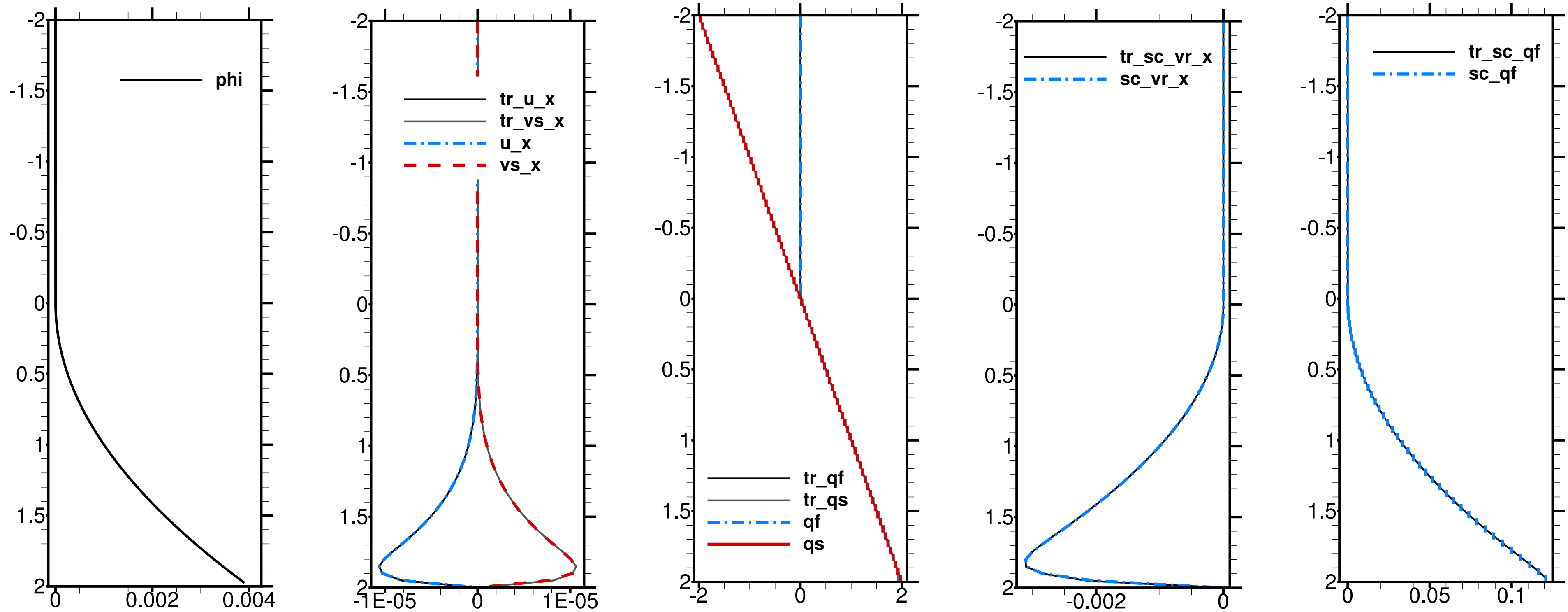
- For $L > z > 0$,

$$u = -v_s = \frac{\phi_+^2}{1 - 4\phi_+}\left(L^{4-r_1}\, z^{r_1} - z^4\right)$$

$$q_f = \frac{1}{1 - 4\phi_+}\left(z - \frac{L^{4-r_1}\, z^{r_1-3}}{r_1 - 3}\right)$$

$$q_s = z$$

- For $-L < z < 0$, $u = v_s = 0$ and $q_s = z$



Porosity ϕ | $u,\ v_s$ | $q_f,\ q_s$ | Scaled $\tilde{v}_r$ | Scaled $\tilde{q}_f$

Approximate relative L^2 errors and convergence rates

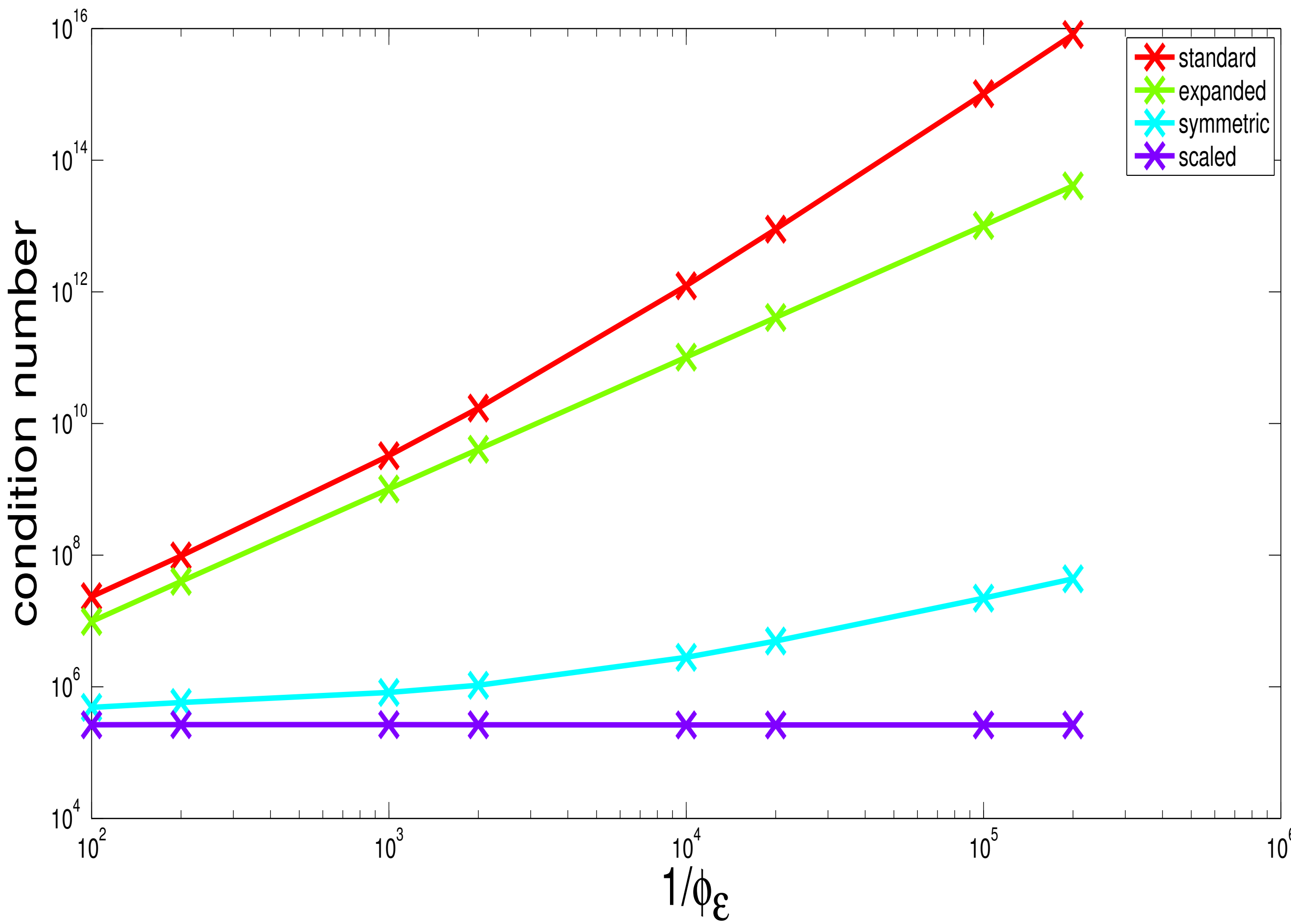
$(h = 4/n,\ L = 2,\ \Theta = 0,\ \phi_+ = 0.001)$

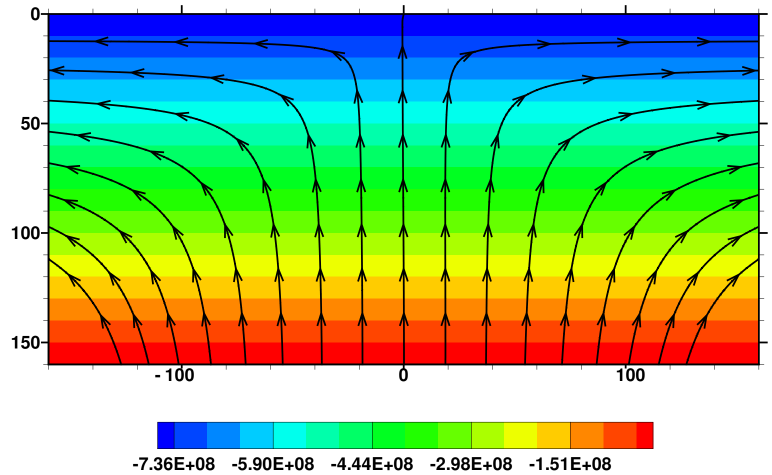
	$\tilde{q}_f$		q		$\tilde{v}_r$		v_s	
n	error	rate	error	rate	error	rate	error	rate
10	1.07e-2	—	6.98e-2	—	8.57e-4	—	3.40e-6	—
20	5.33e-3	1.01	3.49e-2	1.00	3.66e-4	1.23	1.55e-6	1.14
40	2.66e-3	1.00	1.75e-2	1.00	1.17e-4	1.65	5.10e-7	1.60
80	1.33e-3	1.00	8.77e-3	0.99	3.25e-5	1.84	1.43e-7	1.84
160	6.67e-4	0.99	4.46e-3	0.98	1.08e-5	1.59	4.34e-8	1.72
		1		1		2?		2?
11	9.75e-3	—	6.34e-2	—	7.77e-4	—	3.11e-6	—
21	5.07e-3	1.01	3.32e-2	1.00	3.41e-4	1.27	1.44e-6	1.19
41	2.59e-3	1.00	1.70e-2	1.00	1.11e-4	1.67	4.89e-7	1.62
81	1.31e-3	1.00	8.66e-3	0.99	3.18e-5	1.84	1.40e-7	1.84
161	6.63e-4	0.99	4.43e-3	0.98	1.07e-5	1.58	4.30e-8	1.71
		1		1		2?		2?

1-D Compacting Column Condition Number

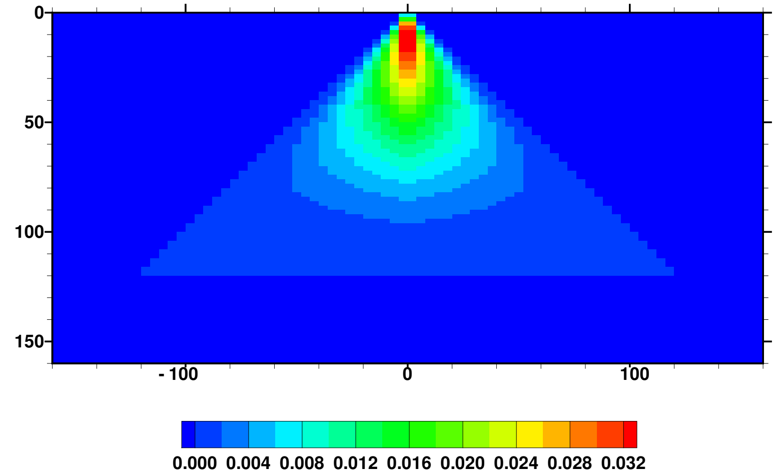
$$\phi(z) = \phi_\epsilon + \phi_0 \begin{cases} 0 & \text{if } z \leq 0 \\ z^2, \ \sqrt{z}, \ 1 & \text{if } z > 0 \end{cases} \qquad \text{Take } \phi_\epsilon \to 0$$

Condition Number of the Linear System

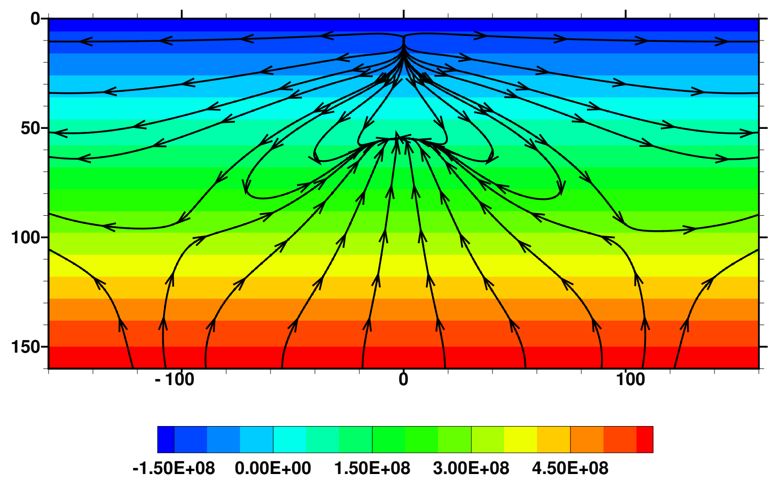




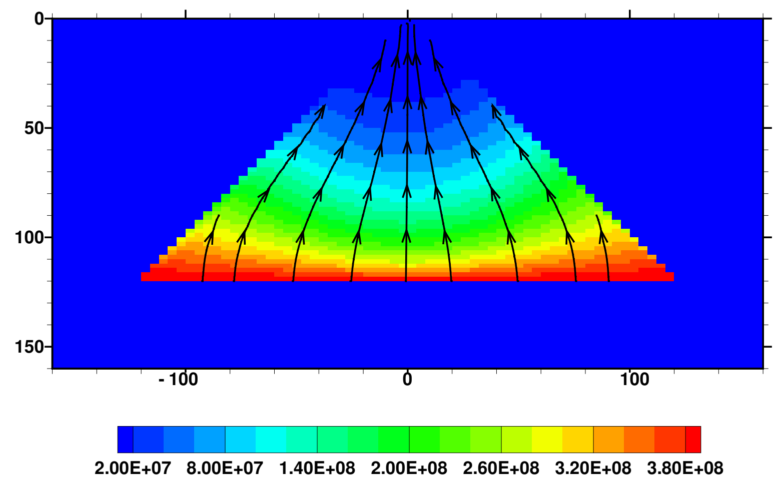
Zero porosity Stokes



Porosity (to 3%)



Stokes System



Darcy System

We used deal.ii to implement the method (with Taylor-Hood elements)

5. Summary and Conclusions

1. We considered a two phase mixture of matrix and fluid melt
   - The equations degenerate as the porosity ϕ vanishes
   - Question: are the equations well-posed?
2. We saw the energy estimates

$$\|\phi^{-1-\Theta}\mathbf{u}\| + \|\phi^{-1/2}\nabla\cdot\mathbf{u}\| + \|\phi^{1/2}q_f\| + \|\mathbf{v}_s\|_1 + \|q\| \leq C$$

   - The potential q_f is uncontrolled in the energy estimates
   - One equation is lost when $\phi = 0$
3. We scaled the Darcy system
   - $\tilde{q}_f = \phi^{1/2}q_f$ and $\tilde{\mathbf{v}}_r = \phi^{1-\Theta}\mathbf{u}$
   - Equations made sense provided $\phi \in L^\infty(\Omega)$ and $\phi^{\Theta-1/2}\nabla\phi \in (L^\infty(\Omega))^d$
4. We defined $H_\phi(\mathrm{div})$ and a normal trace operator, which gave
   - A scaled weak formulation
   - Existence and uniqueness of a solution
   - Continuous dependence on the data

5. We defined a mixed finite element method
   - Proved optimal convergence rates of $\mathcal{O}(h)$
   - Modified the method for local mass conservation
   - Implemented using mass lumping to eliminate $\tilde{\mathbf{v}}_r$
     - Stokes-like system with two pressures
6. Numerical results for a simple degenerate Darcy problem
   - Closed form Euler equation tests showed
     - optimal convergence up to the regularity of the solution
     - Superconvergence
   - Two-dimensional tests showed
     - Superconvergence
     - Importance of $\phi^{-1/2}\nabla\phi \in L^\infty$ (actually, $\phi^{-1/2}\nabla\phi \in L^2$)
     - Mesh need not resolve the one to two phase boundary
7. Numerical mantle dynamics tests showed promise and
   - Captured solution even for discontinuous porosity
   - Low condition numbers for the linear system

When ϕ is reasonable, we have an easy to implement and convergent mixed finite element method

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